Mitul Patil

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LinkedIn | GitHub | Portfolio

Professional Summary

Full Stack Developer (PERN stack) with hands-on experience in production AI integration — RAG pipelines, LLM APIs, vector search, and autonomous agents. Built and deployed three full-stack AI applications from scratch. Shipped a live mobile app on the Google Play Store. Seeking roles at product companies building real AI-powered software.

Education

B.Tech in Computer Science & Engineering June 2022 – May 2026
Parul University, Vadodara, Gujarat
CGPA: 8.59

Technical Skills

Languages: JavaScript (ES6+), Java, Python ,SQL
Frontend: React.js, React Native, HTML5, CSS3, Tailwind CSS, React Router v6, React Hook Form
Backend: Node.js, Express.js, REST API Design, Socket.IO, JWT Authentication, Zod, Bcrypt
Databases: PostgreSQL, pgvector, MongoDB, Supabase, Firebase
AI / LLM: Gemini SDK, RAG Pipeline, AI Agents,Embeddings, Vector Search, LangGraph JS, Tavily API, SSE Streaming, Prompt Engineering
Tools & Deployment: Git, Github ,Vercel, Railway, Postman

Internship Experience

Full Stack Developer Intern – Exotic Infotech Jan 2026 – May 2026

- Translated Figma designs into **10+ pixel-perfect React components**, cutting UI review cycles by **approx. 30%**
- Delivered **3 full-stack MERN modules** (authentication, dashboard, orders), reducing manual admin effort by **approx. 40%**
- Integrated **5+ REST APIs** using Axios with centralized error handling, dropping failed requests to under **2%**
- Optimized Redux state management, improving page load responsiveness across **4 key user flows**

Projects

DocMind – RAG Document Intelligence App Live | GitHub
React, Node.js, Express, PostgreSQL, pgvector, Gemini SDK, , Multer, JWT, Socket.IO, supabase

- Built a **full RAG pipeline from scratch** — PDF parsing, semantic chunking, embedding generation, **pgvector cosine similarity search**, and **streamed LLM responses token-by-token** via SSE to a React frontend
- Implemented **hybrid search** (semantic + keyword), source citations, and per-user **document isolation** with PostgreSQL **row-level security** and prompt injection defense
- Processed large PDFs asynchronously; logged **Gemini API token costs per user**; Perform testing in Postman; deployed on Vercel + Railway + supabase.

ResearchBot – Autonomous AI Research Agent GitHub
React, Node.js, Express, PostgreSQL, pgvector, LangGraph JS, LangSmith ,Gemini SDK, Tavily API

- Built an **autonomous research agent LangGraph JS** state machine — plan, web search (Tavily), URL fetch, synthesize, and save nodes with **conditional edges** and human-in-the-loop approval, with **LangSmith tracing** for workflow observability.
- Implemented **agent memory** using pgvector semantic search over past reports, **structured JSON output** with Gemini responseSchema, and **LangSmith evaluation** to monitor and improve agent response quality.
- Implemented **real-time step streaming** to React via SSE, added prompt-injection defenses for AI agents, and optimized PostgreSQL queries using **EXPLAIN ANALYZE**.

Greenlit – AI-Powered Freelancer Decision Assistant GitHub
React.js, Node.js, Express.js, PostgreSQL, LLM API, TipTap, JWT

- Automated **Proposal, Scope, and Risk document generation** using an LLM, reducing creation time by **approx. 70%**
- Engineered a **TipTap rich-text editor** with **8s auto-save** and one-click PDF/DOCX export; built a **JWT-authenticated Discovery Portal** with client Q&A workflow and shareable project workspace

Mr. Dhobi – On-Demand Laundry Mobile App Live (Play Store) | GitHub
React Native, TypeScript, Firebase, Razorpay

- Shipped a **live Android app on Google Play Store** with **OTP authentication**, push notifications, and **Razorpay** payment integration
- Implemented a **role-based admin panel** with **10+ operational modules** managing orders, delivery, and user roles; companion website scored **98/100 Lighthouse performance score**

Certifications

Full Stack Web Developer – Udemy (2025)